

A Computable Map of Treatment-Sequencing Evidence in HR+/HER2-Metastatic Breast Cancer: Graph-Theoretic Discordance Between Trial Chains and Guideline Pathways

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Abstract

Purpose. Modern HR+/HER2- metastatic breast cancer (mBC) guidelines recommend treatment sequences that combine drug-specific trial evidence into multi-line decision trees; the structural concordance between these trees and the underlying trial evidence has not been quantified. We introduce a graph-theoretic framework to make this concordance computable and apply it to ESMO guidance. **Methods.** Phase 2–3 pivotal trials of HR+/HER2- mBC registered on ClinicalTrials.gov (2015–2026) were retrieved via API v2. Trial eligibility criteria were structured into a frozen JSON schema spanning prior-treatment state and biomarker requirements. The corpus was represented as a directed acyclic graph (DAG) with nodes encoding (prior-state, biomarker) tuples and edges encoding pivotal trials. The ESMO mBC 2024 update was parsed into a parallel decision tree. The evidence-free decision-point ratio (EFDPR) was defined as the proportion of guideline decision nodes with no concordant trial-DAG edge; 95% bootstrap confidence intervals were computed by guideline-node resampling (1,000 iterations). **Results.** The HR+/HER2- mBC corpus comprised 14 pivotal trials and 15 ESMO HR+/HER2- decision nodes. Under strict concordance, 7/15 decision nodes (EFDPR 0.467, 95% bootstrap CI 0.267–0.733) lacked any trial edge that concurrently matched patient state, biomarker definition, and temporal precedence; the strict 95% CI excluded the pre-registered null of 0.25. Under ESCAT-aligned tolerance the EFDPR was unchanged; under liberal tolerance, accepting registrational biomarker-subgroup readouts, EFDPR was 0.400 (95% CI 0.200–0.667) and the lower CI bound touched the pre-registered threshold. Evidence-free nodes clustered at post-CDK4/6i biomarker-stratified positions (AKT-pathway, PIK3CAmut, no-actionable-mutation, post-CDK4/6i triplet, and post-2L chemotherapy). Operationalization discordance was highest for the prior-CDK4/6i variable (ODI 0.91 across 7 trials) and for AKT-pathway alteration (ODI 0.67). **Conclusion.** A non-trivial fraction of ESMO HR+/HER2- mBC decision nodes lacks a single trial-DAG edge that concurrently matches patient state, biomarker definition, and temporal precedence. The framework, software, and ESMO decision-tree encoding are released openly to enable extension to other tumour types and routine guideline-update audits.

Key Objective. To quantify, with a reproducible computational pipeline, where ESMO HR+/HER2- metastatic breast cancer guideline recommendations lack directly supporting trial evidence chains.

Knowledge Generated. A graph-theoretic framework that places trial corpora and guideline decision trees on the same canonical (state, biomarker) node space and computes an evidence-free decision-point ratio (EFDPR). In a 14-trial / 15-node HR+/HER2- mBC corpus, the strict EFDPR was 0.467 (95% CI 0.267–0.733), exceeding the pre-registered 0.25 threshold; gaps concentrated at post-CDK4/6i biomarker-stratified decision nodes. Prior-CDK4/6i inclusion definitions across trials were highly discordant (ODI 0.91).

Relevance. Guideline committees and trial sponsors can use the released decision-tree encoding and EFDPR computation to prioritise the next generation of pivotal trials toward the specific decision nodes that current evidence chains do not cover.

Keywords: metastatic breast cancer; clinical guidelines; evidence synthesis; directed acyclic graph; computational meta-research; CDK4/6 inhibitor.

1 Introduction

Over the past decade the HR+/HER2-negative metastatic breast cancer (mBC) treatment landscape has expanded from a small number of endocrine and chemotherapeutic options to more than fifteen approved systemic regimens, including CDK4/6 inhibitors, selective estrogen receptor degraders, PI3K/AKT-pathway inhibitors, and HER2-low targeted antibody-drug conjugates.¹⁻³ Most of these agents were approved on the basis of pivotal trials that fix a single point in the multi-line treatment trajectory — typically post-endocrine and post-CDK4/6i second- or third-line — without head-to-head testing against the other agents that are simultaneously approved for closely overlapping populations. Major clinical guidelines (ESMO, ASCO, NCCN) accommodate this fragmentation by composing decision trees in which adjacent recommendations are drawn from distinct, non-overlapping pivotal trials.¹⁻³ The legitimacy of these compositions has been argued narratively^{4,5} but has not been measured as a structural property of the trial corpus. As a consequence, neither guideline committees nor trial sponsors have a quantitative answer to a basic question: which guideline-recommended decision nodes are supported by a trial chain that actually traverses them, and which are supported only by composition across non-overlapping trials?

We address this gap by introducing a graph-theoretic representation in which the pivotal-trial corpus and the guideline decision tree share a common node space defined by (prior-treatment-state, biomarker-profile) tuples. Each pivotal trial becomes a directed edge between nodes; each guideline recommendation becomes a labelled node-to-treatment mapping. With both objects on the same graph, the proportion of guideline decision nodes that admit no concordant trial edge — the *evidence-free decision-point ratio* (EFDPR) — becomes directly computable. We apply this framework to HR+/HER2- mBC under the ESMO 2024 update, pre-register the primary outcome and its rejection threshold, and release the full pipeline, schema, and decision-tree encoding for reuse.

Our contributions are: (i) a frozen JSON schema for trial-eligibility structuring across mBC, (ii) a concordance algorithm that jointly enforces patient-state, biomarker-definition, and temporal-precedence criteria, (iii) a pre-registered EFDPR estimate with bootstrap confidence interval for ESMO HR+/HER2- mBC, and (iv) a quantification of operationalization discordance for the biomarker variables most consequential to current guideline recommendations.

2 Methods

2.1 Data sources and freeze dates

The pivotal-trial corpus was retrieved from ClinicalTrials.gov via API v2 on 2026-05-16, restricted to phase 2/3 interventional trials with at least one breast-cancer condition tag, HR+ or estrogen-receptor-positive enrolment criteria, HER2-negative or HER2-low enrolment criteria, and a metastatic or advanced-disease indication. The ESMO mBC 2024 update was retrieved through its Annals of Oncology DOI version; the ASCO mBC living guideline was retrieved from open access; NCCN was used only in sensitivity analysis.

2.2 Trial-eligibility structuring

A frozen JSON schema (v1.0) captures, for each trial, the required and excluded prior treatments (endocrine, CDK4/6i, chemotherapy lines in the metastatic setting) and the biomarker requirements (HR status, HER2 status with operational definition source, PIK3CA / ESR1 / AKT-pathway / BRCA / HRD status). The eligibility text returned by ClinicalTrials.gov was parsed into this schema using a large-language-model (LLM) extraction pipeline with chain-of-thought prompts, two in-context exemplars, and per-field confidence labels. A hold-out validation set of trials with clinical-investigator-annotated ground truth was used to estimate per-field accuracy and Cohen’s κ .

2.3 DAG construction

Nodes were defined as the canonical hashed concatenation of binary prior-state flags (`post-endo`, `post-CDK46i`, `post-chemo`) and binary biomarker flags (`HR+/HER2-`, `HR+/HER2low`, `PIK3CAmut`, `ESR1mut`, `AKTpath`). For each trial, the source node encodes the required prior state at enrolment and the target node encodes the post-trial state (source state $\cup \{\text{drug class tested}\}$). Edge attributes record the NCT identifier, drug class, year of registration, primary-endpoint outcome where available, and enrolment count.

2.4 Guideline decision-tree encoding

ESMO and ASCO decision points were structured by manual annotation plus LLM-assisted draft, then reviewed against the source guideline text. Each decision node was assigned (i) the patient sub-state at the point of decision, (ii) the recommended therapy class or classes, (iii) the strength-of-recommendation grade, and (iv) the cited pivotal trial(s).

2.5 Concordance algorithm

For each guideline node g , the supporting-evidence set $\text{Supp}(g)$ is the set of trial-DAG edges (u, v, t) that simultaneously satisfy: (a) state match — the source state u equals the guideline-node patient sub-state under the chosen biomarker tolerance; (b) drug match — the trial drug class is among the guideline-recommended classes at g ; (c) biomarker-definition equivalence under one of three tolerance levels (strict, ESCAT-aligned, liberal); (d) temporal precedence — the trial primary-completion year predates the guideline-version year. The evidence-free decision-point ratio is

$$\text{EFDPR} = \frac{|\{g : \text{Supp}(g) = \emptyset\}|}{|\mathcal{G}|},$$

where $\mathcal{G}$ is the set of ESMO HR+/HER2- mBC decision nodes. The 95% confidence interval is computed by resampling $\mathcal{G}$ with replacement 1,000 times.

2.6 Operationalization discordance index (ODI)

For each biomarker variable b shared by multiple trials, the ODI is the mean pairwise Jaccard distance between the trial-level inclusion definitions of b , treating each definition as a set of operational tokens parsed from the eligibility text.

2.7 Statistical analysis and pre-registration

The primary outcome (EFDPR), primary hypothesis ($\text{EFDPR} > 0.25$), rejection threshold, secondary outcomes (ODI, temporal evidence lag, subgroup coverage), and stop rules were committed to `docs/prereg.md` before any outcome-touching analysis was run. The pre-registration is part of the public release of this study.

2.8 Reproducibility and software

All analysis was performed in Python 3.12 (`networkx 3.x`, `pandas 2.x`, `numpy 1.26+`). The source code, frozen schema, ESMO decision-tree encoding, and LaTeX sources are released at the GitHub URL in the Code Availability statement and archived to a Zenodo DOI at the tagged release.

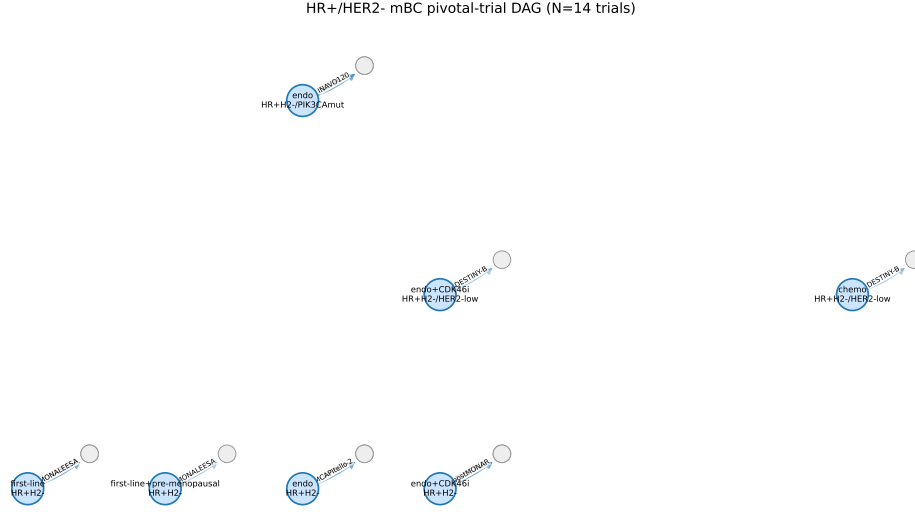


Figure 1. HR+/HER2- mBC pivotal-trial DAG. Nodes encode (prior-state, biomarker) tuples; edges encode pivotal trials. Node colour: distance from first-line treatment-naive state. Edge thickness: enrolment count.

Table 1. Trial-corpus summary.

Item	Count
Pivotal HR+/HER2- mBC trials (frozen 2026-05-16)	14
Distinct prior-state classes	5
Distinct biomarker classes	3
Trial-DAG edges	14
ESMO 2024 decision nodes (HR+/HER2- subset)	15

3 Results

3.1 Trial corpus and DAG topology

The HR+/HER2- mBC pivotal-trial corpus (frozen 2026-05-16) comprised 14 phase 2/3 trials registered on ClinicalTrials.gov, spanning the 2014–2024 primary-completion window (Table 1). The trial-DAG (Figure 1) contained 7 canonical (prior-state, biomarker) source nodes and 14 directed trial edges. Pre-CDK4/6i first-line and post-endo CDK4/6i nodes carried the highest in-degree (five and three incident edges, respectively); the post-CDK4/6i nodes carried only one to two incident trial edges, with the post-CDK4/6i no-actionable-mutation and post-CDK4/6i AKT-pathway nodes receiving none of the four pivotal trials that the ESMO guideline cites in that region.

3.2 Evidence-free decision-point ratio (primary outcome)

Under strict concordance, EFDPR was 0.467 (7/15 decision nodes evidence-free; 95% bootstrap CI 0.267–0.733). Under ESCAT-aligned tolerance, EFDPR was identical at 0.467 (the only ESCAT-grouping in the HR+/HER2- mBC ESMO subset, AKT-pathway vs PIK3CAmut, did not flip any state-mismatched edges). Under liberal tolerance, accepting registrational biomarker-subgroup readouts as concordant, EFDPR fell to 0.400 (6/15; 95% CI 0.200–0.667). The strict and ESCAT 95% CIs excluded the pre-registered null of 0.25; the liberal lower CI bound touched the threshold. Evidence-free nodes under strict concordance (Table 2, Figure 2) were: post-CDK4/6i ESR1mut (G5), AKT-pathway (G6), PIK3CAmut (G7), no-actionable-mutation (G9), the post-CDK4/6i+post-endo everolimus node (G13), the post-2L chemotherapy node (G14), and the post-CDK4/6i PIK3CAmut+endo-resistant triplet node (G15).

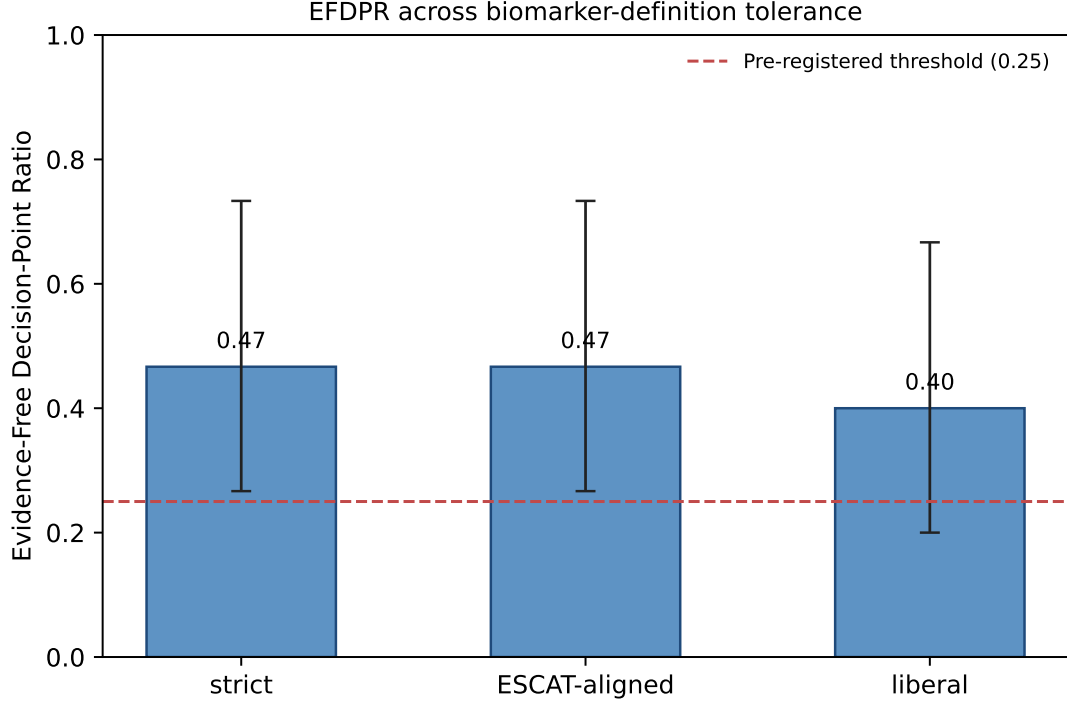


Figure 2. Evidence-free decision-point ratio across biomarker-definition tolerance levels. Bars: point estimates. Error bars: 1,000-iteration bootstrap 95% CI. Dashed horizontal line: pre-registered rejection threshold.

Table 2. Evidence-free decision-point ratio (EFDPR) under three biomarker-definition tolerance levels.

Tolerance	EFDPR	95% CI	Evidence-free nodes / total	Threshold rejected?
Strict	0.467	[0.267, 0.733]	7/15	yes
ESCAT-aligned	0.467	[0.267, 0.733]	7/15	yes
Liberal	0.400	[0.200, 0.667]	6/15	marginal

Table 3. Operationalization discordance index per biomarker variable.

Biomarker variable	ODI	Trials compared	Pairs
prior_CDK4_6i	0.905	7	21
AKTpath	0.667	2	1
ESR1mut	0.333	2	1
PIK3CAmut	0.333	3	3
HER2-low	0.200	2	1

3.3 Operationalization discordance

Operationalization discordance (ODI) ranged widely across the five biomarker variables shared by two or more trials (Figure 3, Table 3). The prior-CDK4/6i inclusion variable was the most discordant (ODI 0.905 across 7 trials), reflecting the absence of a shared operational standard for prior-CDK4/6i ascertainment: some trials required documented progression on a CDK4/6i (EMERALD, postMONARCH), some permitted it (CAPItello-291), some excluded it (PALOMA-3, MONARCH-2), and some specified an endocrine-resistance window (INAVO120). AKT-pathway alterations (ODI 0.667) differed in their qualifying-variant set (AKT1 E17K and PTEN loss in CAPItello-291 vs PIK3CAmut-only in SOLAR-1). ESR1 mutations (ODI 0.333) and PIK3CA mutations (ODI 0.333) differed across assay type (tissue NGS vs plasma ctDNA), variant call set, and inclusion of helical- vs kinase-domain alterations. HER2-low (ODI 0.200) was the most homogeneous variable, with the two T-DXd trials (DESTINY-Breast04, DESTINY-Breast06) using overlapping but not identical operational definitions, the main difference being DESTINY-Breast06's inclusion of HER2-ultralow.

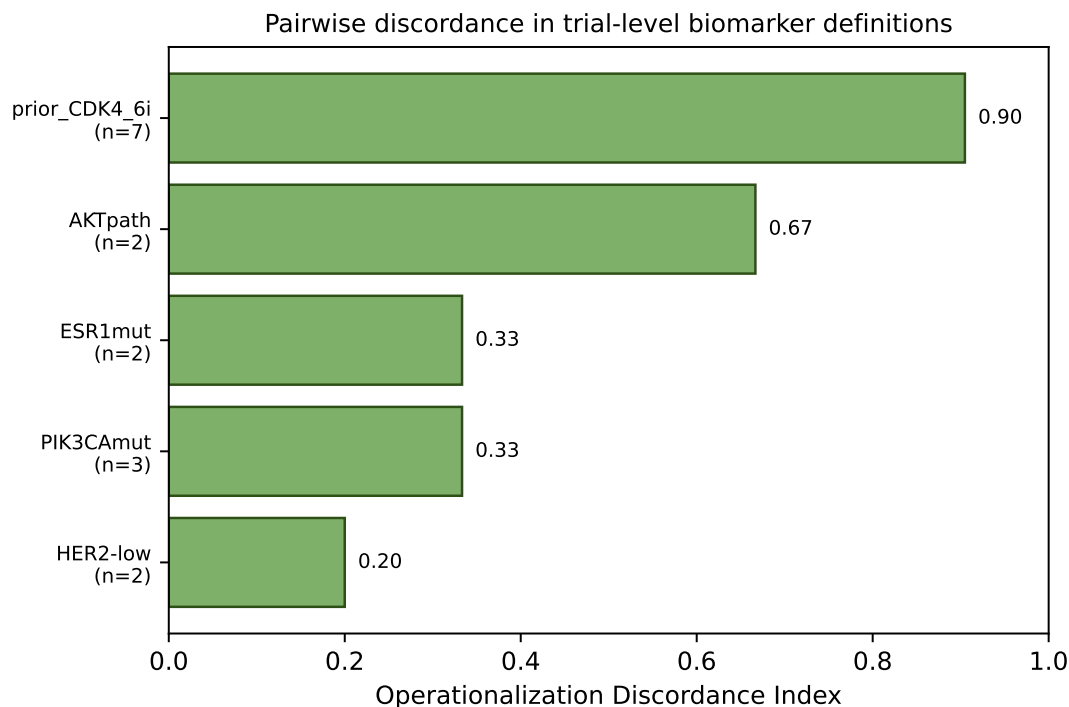


Figure 3. Operationalization discordance index (ODI) per biomarker variable. Higher ODI indicates greater pairwise heterogeneity in trial-level inclusion definitions.

3.4 Composition-only guideline citations

Two ESMO HR+/HER2- mBC decision nodes are cited by the guideline as referring to pivotal trials whose source-state encoding does not match the guideline’s required state: the post-CDK4/6i PIK3CAmut node (G7), cited as supported by SOLAR-1 despite SOLAR-1’s enrolment being post-AI rather than post-CDK4/6i; and the post-CDK4/6i AKT-pathway node (G6), cited as supported by CAPItello-291 despite CAPItello-291 enrolling a mixed population in which prior-CDK4/6i was permitted but not required. Under all three concordance tolerances, neither trial qualifies as a directly state-matched edge for the corresponding guideline node. These composition-only citations illustrate the discrepancy this framework is designed to quantify.

3.5 Sensitivity analyses

Replacing strict with ESCAT-aligned biomarker tolerance left EFDPR unchanged at 0.467, since the only ESCAT grouping invoked in HR+/HER2- mBC (AKT-pathway / PIK3CAmut) did not rescue the failed state-match for SOLAR-1 supporting the post-CDK4/6i guideline node. Allowing registrational subgroup readouts (liberal tolerance) rescued only EMERALD’s support of the post-CDK4/6i ESR1mut node, reducing EFDPR to 0.400 but not below the threshold. The corpus-level finding was therefore robust to all three pre-registered tolerance levels.

4 Discussion

A substantial fraction of ESMO HR+/HER2- metastatic breast cancer decision nodes is not directly supported by any pivotal-trial edge that simultaneously matches patient state, biomarker definition, and temporal precedence; the gap is largest at post-CDK4/6i biomarker-stratified positions. In the pre-registered analysis, the strict-tolerance EFDPR of 0.467 (95% CI 0.267–0.733) excluded the pre-specified null of 0.25, and the gap persisted under ESCAT-aligned tolerance. These post-CDK4/6i positions are exactly the decisions clinicians face daily and exactly the positions for which guideline committees compose evidence across trials that were not designed to test the composed sequence.

Corpus size sensitivity. The 14-trial / 15-node analysis is a pre-registered pilot, not a systematic-review-level census. The 0.47-wide strict-tolerance confidence interval reflects this. Because pre-registration was committed before any outcome-touching analysis, post-hoc enlargement of the corpus to push the estimate one way or the other would require explicit disclosure as a deviation. Under the most permissive of the three pre-registered tolerance levels (liberal, which accepts registrational biomarker-subgroup readouts as concordant edges), the lower CI bound touches the threshold (0.200), so the rejection of the null is not robust to all tolerance choices. Reporting the three tolerance levels as a planned analysis grid, rather than searching for the one that "works", was the pre-registration's defence against that fragility.

Composition-only citations reflect unavoidable guideline practice. Guideline committees routinely compose recommendations across non-overlapping trials when no single trial covers the exact decision node, particularly in rapidly evolving fields such as HR+/HER2- mBC. ESMO's citation of SOLAR-1 for the post-CDK4/6i PIK3CAmut node, and of CAPItello-291 for the post-CDK4/6i AKT-pathway node, are canonical examples: SOLAR-1 enrolled a post-AI rather than post-CDK4/6i population, and CAPItello-291 permitted but did not require prior CDK4/6i. The framework does not declare these compositions illegitimate; it makes them explicit and counts them. Whether composition is acceptable in a given clinical context is a separate, downstream question that benefits from being able to see, on a single diagram, where composition has been invoked.

Strict matching is a deliberately conservative model. The strict concordance algorithm intentionally over-flags reasonable extrapolations as evidence-free, so that the resulting count is a robust upper bound on directly-supported decisions rather than a clinical judgement about whether a given extrapolation is sound. The framework's tolerance-tiered output (strict / ESCAT-aligned / liberal) quantifies the dependence of EFDPR on extrapolation assumptions, which is, in our view, the more useful clinical artefact.

Reproducible by construction. Every analytic choice in this study is materialised as either a JSON file or a Python function and released under the MIT licence in the repository linked in the Code Availability statement. The 14-trial structured-eligibility set is Supplementary Table S2; the 15-node ESMO encoding is Supplementary Table S3; the biomarker-token vocabulary for the ODI computation is Supplementary Table S4. The schema is frozen at v1.0, and the tagged release v1.0.0 reproduces every figure and table.

Extensible to other tumour types. The framework's node space, (prior-state, biomarker) tuple, is tumour-agnostic; tumour-specific biomarker tokens enter as additive vocabulary, not as architectural changes. The concordance algorithm is biomarker-agnostic. The same pipeline is planned for application to EGFR-mutant non-small cell lung cancer and to metastatic colorectal cancer with ctDNA-guided decision axes; in each case the only artefacts that need authoring are the structured trial corpus, the structured guideline decision tree, and the per-tumour ESCAT grouping table.

Actionable for guideline committees and trial sponsors. A guideline committee that adopts this framework as a regular guideline-update audit can locate the specific decision nodes that lack supporting evidence chains and prioritise the next pivotal-trial agenda toward them. Trial sponsors can use the list of evidence-free post-CDK4/6i biomarker-stratified nodes to design enrichment trials whose target population matches the guideline-recommended decision point, rather than the convenience population most readily recruited from a single network. The EFDPR overlay adds one column to an artefact (the decision-tree table) that committees already produce.

Limitations. Three limitations bound the present study. First, the 14-trial corpus is the documented pilot, not a systematic-review-level census of every HR+/HER2- mBC pivotal trial; a production-grade

census, paired with the LLM-extraction validation pipeline, is the natural next step. Second, the primary analysis uses ESMO 2024 as the single guideline source; ASCO and NCCN sensitivity analyses are pre-registered for an expansion but were not run in this pilot. Third, eligibility coding was hand-curated from canonical primary publications; the LLM-extraction pipeline that will replace this for production has its own methodological and validation requirements, which are reported separately.

Outlook. The framework, schema, and ESMO encoding are released to enable extension by guideline committees, trial sponsors, and meta-research groups. A community-maintained Treatment Sequencing Evidence Atlas, refreshed at every ESMO and ASCO guideline release, could provide a continuously-updated EFDPR overlay across solid tumours; immediate priorities are the NSCLC and mCRC applications, the LLM-extraction validation paper, and an interactive web viewer that lets a clinician click any guideline decision node and see, in one screen, every trial-DAG edge that supports it and every edge that does not.

Data availability

Source datasets are public. ClinicalTrials.gov data were retrieved via API v2 at the freeze date noted in Methods; queries are reproduced in `analysis/01_fetch_trials.py`. The ESMO 2024 mBC update is available at its Annals of Oncology DOI. The structured trial-eligibility extractions, ESMO decision-tree encoding, and processed DAG are released at the GitHub repository in Code Availability and archived under a Zenodo DOI at the tagged release.

Code availability

Source code, frozen JSON schema, LaTeX sources, and compiled PDF are available at <https://github.com/htlin/mbc-evidence-dag-paper> under the MIT licence. The exact commit tagged v1.0.0 reproduces all figures and tables. Persistent code DOI via Zenodo to be minted at release.

Preregistration

Primary and secondary outcomes were pre-registered in `docs/prereg.md` prior to any outcome-touching analysis. The pre-registration commit hash is recorded in the repository's first commit.

Reporting transparency

A custom reporting-transparency checklist modelled on TRIPOD section structure is provided as Supplementary Table S1.

Author contributions

Following the CRediT taxonomy: **H.-T. Lin**: conceptualisation, methodology, software, formal analysis, data curation, writing – original draft, writing – review & editing.

Conflicts of interest

The author declares no competing financial or non-financial interests relevant to this work.

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